

Can TLC trust its wheelchair-accessible 10-minute numbers?

PROBLEM

TLC cannot verify monthly whether Uber and Lyft serve wheelchair users on time: the evidence is split across 21 M-row trip files and control reports, and timestamp errors inflate compliance.

KPI

WAV 10-min service rate: share of wheelchair-accessible (WAV) requests with the vehicle *on scene* < 600 s after the request, per dispatcher-month. Target ≥ 90% (35 RCNY §59B-17(f)(3)).

DECISION SUPPORTED

TLC accessibility compliance: open a compliance review or a data-correction request; where and when to target WAV supply; which months are safe to publish.

Headline. Uber meets the standard (94.3% across Apr/May/Jul 2026): *proven* in April, an *estimate* in May/July, where the bounds straddle 90%. The margin is shrinking (95.8 → 94.8 → 92.2%). **Lyft's compliance cannot be determined:** 58–61% of its WAV records have the vehicle arriving *before* the request (raw median –6.4 min). Unvalidated, Lyft looks compliant (92.7%). Silently filtered, it looks non-compliant (81.0%). The pipeline **withholds** the number and flags a data-correction request. **June 2026 was refused:** the file is 2.14% short of TLC's own monthly report.

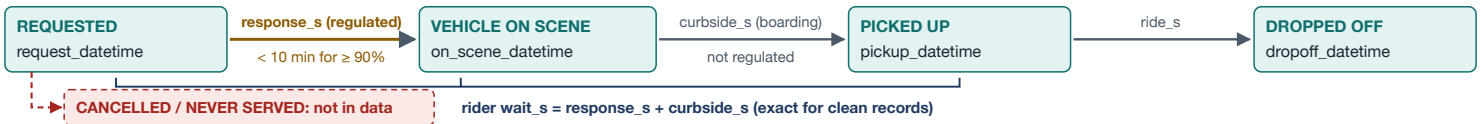
- 1 PROBLEM WORTH SOLVING**
 - A regulated, enforceable standard (notice + 30 days to comply) for a vulnerable rider group, measured on **data the dispatchers submit themselves**.
 - TLC's FY23 accessibility report, recommendation #1: "Add FHV WAV service metrics to the TLC Dashboard." Today it is an annual PDF.
- 2 WHY IT MATTERS (AND WHY THIS TRACK)**
 - The 2025 rule made Uber/Lyft report on-scene time "so that TLC can better compare WAV and non-WAV wait times". M3/M4 do exactly that.
 - Only High-Volume FHV records carry request **and** on-scene times. Users: TLC compliance (primary), the TLC data team, Uber/Lyft WAV ops.

3 SOURCE MAP — QUESTION → INFORMATION → SOURCE → FIELDS → METRIC

#	Source · owner	Retrieval mode	Grain	Fields used → metric	How completeness is proven
S1	HVFHV trip records · TLC (submitted by Uber/Lyft, §59D-14)	File: Parquet over HTTPS, ~0.5 GB/month	1 row = 1 completed trip; no trip ID	request / on_scene / pickup time, wav flags, licensee → M1–M5	bytes = Content-Length, SHA-256; rows read = Parquet footer; all days present
S2	Taxi zone lookup · TLC	File: CSV over HTTPS	1 row = 1 zone (265)	PULocationID → Borough → M5	schema contract; LocationID unique (safe join key)
S3	FHV Base Aggregate Report · TLC on NYC Open Data	API: Socrata SODA JSON, paginated	base × month (UBER, LYFT rows)	total_dispatched_trips → licensee control	server count(*) = rows retrieved over ordered pages
S4	Monthly industry report · TLC	File: CSV over HTTPS	licence class × month, trips <i>per day</i>	trips/day × days → timely fallback control	month row present; ±0.5% tolerance (S4 is tabulated slightly differently)

Gap: WAV requests never served or cancelled exist in no public source. Every rate is on completed trips, so it is an **upper bound** on regulatory compliance. · **Grain mismatch handled:** trips are aggregated to licensee × month (S3) and class × month (S4) before comparing. S3/S4 come from the same submissions, so they prove the file was *received* complete, not that every trip was *submitted*.

4 WORKFLOW AND DATA MODEL



Entities: dispatcher (licensee → base B03404 Uber / B03406 Lyft) · trip · pickup zone → borough · vehicle only via wav_match_flag (no ID). **Intervention:** WAV dispatch duty, TLC notice. **Outcome:** on scene < 10 min · rider wait · gap vs non-WAV. **Model (DuckDB):** fct_wav_trip PK (month, source_row_id) = 1 WAV-requested trip, quarantined rows kept with reasons · ref_licensee / ref_zone PKs · agg_segment_month PK (month, level, licensee, WAV, borough, band) over all 21 M trips → views M1–M5. **Joins are safe:** each is a LEFT JOIN onto a reference PK (many → 1), so it cannot drop or multiply trips; unmatched keys fail REF01/02. Tested: dispatcher totals = fact rows = ∑ cells.

5 KEY EVIDENCE — WHAT VALIDATION CHANGES

Month	Dispatcher	WAV requests	Coverage (M2)	Naive (no validation)	Clean-record rate (= M1 if measurable)	Bounds*	Verdict
2026-04	Uber	42,326	95.6%	96.0%	95.8%	91.6–96.0	MEETS proven
2026-04	Lyft	22,920	40.0%	94.1% ✓	85.2% ✗ (not decision-grade)	34.0–94.1	NOT MEASURABLE
2026-05	Uber	42,486	94.9%	95.1%	94.8%	89.997–95.1	MEETS estimate
2026-05	Lyft	26,682	38.6%	92.7% ✓	81.0% ✗ (not decision-grade)	31.3–92.7	NOT MEASURABLE
2026-07 ^P	Uber	43,573	94.6%	92.6%	92.2%	87.2–92.6	MEETS estimate
2026-07 ^P	Lyft	23,176	42.0%	93.2% ✓	83.8% ✗ (not decision-grade)	35.2–93.2	NOT MEASURABLE

*Bounds: every quarantined record counted as a fail ... as a pass. **Verdict:** both bounds on one side of 90% → **proven**; bounds straddle and coverage ≥ 90% → **estimate** (quarantined records treated as out of scope); otherwise **NOT MEASURABLE**. ✓/✗ = the verdict that number alone would imply. ^P = provisional. For Uber, M1 is the clean-record rate; validation decides when it may be used. Source: outputs/metrics/m1_m2_kpi_dispatcher_month.csv.

Why this problem (chosen before any code): Track A would need invented data · 311 tickets: only created/closed times · yellow taxi: no request or on-scene time → **HVFHV + the WAV rule:** a published regulatory KPI, a 4-event workflow, independent controls and verifiable defects.

6 RETRIEVAL — COMPLETENESS PROVEN

Month	Trips in file	vs S3 API (Uber/Lyft)	vs S4 report (all)	Result
2026-04	20,995,953	exact / exact	+63 (+0.0003%)	FINAL
2026-05	22,125,744	exact / exact	+114 (+0.0005%)	FINAL
2026-06	20,775,868	unpublished	-454,232 (-2.14%)	REFUSED
2026-07	20,921,249	unpublished	+62 (+0.0003%)	PROVISIONAL

- June:** FIL07 flags **Lyft on 8–10 June** (113.5–117.2 K/day vs a 187.7 K median). Lyft stays low to 14 June; its deficit vs the same weekdays ≈ 489 K ≈ the gap. *Inference:* missing submissions. Evidence is kept in `validation/2026-06/`; nothing was published.
- Files:** `.part` → wire bytes = Content-Length → SHA-256 → atomic rename; rows read = Parquet footer. **API:** `server count(*)` = rows paged with a total `$order`. Raw is preserved with manifests.
- Real-data lesson:** nyc.gov gzips S4, so Content-Length is the compressed size and the first byte check refused the file. Fixed with identity encoding and a wire-byte comparison.

7 VALIDATION — 13 RULES, NOTHING SILENTLY FIXED

Issue	Evidence (Apr / May / Jul 2026)	Decision
Lyft WAV: on scene before request (TMP02)	60.0 / 61.4 / 58.0% ; raw median -6.4 min; one base; separate record path; whole-minute requests 2.1–2.5% (≈ chance)	quarantine; KPI withheld
Uber: same rule	4.4 / 5.1 / 5.4%; 99.95–100% whole-minute requests (valid rows 1.8%)	<i>pre-booked signature</i> ; bounds
Dictionary: on-scene "WAV only"	0.00% null for all 20.8–22.1 M trips/month	trust data → M3/M4
On-scene = pickup to the second	6.4% Uber non-WAV; 21.9% at LGA/JFK vs 5.8% elsewhere	WARN; caveat on M4
Waits > 60 min	≤ 0.005% of any segment	kept as failures

Rules: schema · referential · temporal (request ≤ on-scene ≤ pickup < dropoff) · domain · grain (exact duplicates; no trip ID). Duplicates, out-of-month rows, bad keys: 0 every month; a synthetic month proves each rule fires. Profile: exact distinct counts + raw response distribution per segment.

8 FIVE METRICS TIED TO THE KPI (UBER SHOWN; LYFT WITHHELD)

Metric	Definition	Apr	May	Jul	So what / decision
M1 10-min rate	clean WAV requests with on-scene – request < 600 s ÷ clean WAV requests (+ bounds, verdict)	95.8%	94.8%	92.2%	meets (Apr proven, May/Jul estimate); YTD 94.3% → monitor
M2 coverage	WAV records passing every ERROR rule ÷ all WAV records (policy line 90%)	95.6%	94.9%	94.6%	Lyft 38.6–42.0% → data-correction request, no verdict
M3 P90 gap	P90 response WAV – P90 non-WAV, same dispatcher (P90 < 10 ⇔ 90% under 10)	-0.92 m	-1.05 m	-0.22 m	parity in the tail only; P50 ~1 min slower
M4 curbside share	extra mean wait (WAV – non-WAV) that accrues after arrival; exact since wait = response + curbside	84.3%	85.4%	79.1%	~2.3 of 2.8 extra min are outside the 10-min clock
M5 below-target demand	WAV requests in cells (borough × time band, n ≥ 100) whose 95% Wilson interval is below 90% ÷ all judged	1.4%	1.7%	7.2%	Queens 20–24 every month (Jul 70.0% vs non-WAV 81.6%); July adds Queens/Bronx night windows

9 DEPENDABILITY — INGEST → PROFILE → VALIDATE → RECONCILE → MODEL → METRICS

What happens if...	Behaviour (every row has a test; 54 total)	What happens if...	Behaviour
it runs twice	month DELETE + INSERT in one transaction, PKs enforce grain → byte-identical outputs	the schema changes	missing or retyped column → stop; new column → WARN only
a download truncates / 5xx	retry ×3 with backoff; <code>.part</code> never renamed into place	raw tampered / control mismatch / refresh fails	month stopped, last good kept ; refused-month evidence written
the month isn't published (404)	fail fast, no retries; other months still run; exit code 1	duplicates / impossible rows	quarantined with rule IDs; verdict from the bounds
the API is down / paginates short	reuse preserved pages, else PROVISIONAL; short pages detected	no network	<code>--offline</code> rebuilds identically from preserved raw

Two rebuilds of all 4 months (84 M rows) from an empty warehouse give **byte-identical outputs** (only `last_run.json` differs) · per-run logs · append-only `pipeline_run` audit.

10 KNOWN · UNKNOWN · ASSUMPTION · LIMITATION

KNOWN Rule: ≥ 90% under 10 min; passenger cancels within 10 min excluded. TLC measures to vehicle arrival. Apr/May reconcile exactly; the June file is -2.14%. 0 duplicates.	UNKNOWN Unserved or cancelled WAV requests. The cause of Lyft's defect. Whether pre-booked rides fall under the rule. Whether June's gap is Lyft (<i>inferred</i>).	ASSUMPTION Population = completed trips with <code>wav_request_flag=Y</code> ; dispatcher = license; "under" = < 600 s; month = pickup month; an "estimate" treats quarantined rows as out of scope (Uber: pre-booked signature).	LIMITATION Rates are upper bounds (completed trips only). Controls prove the file was received, not that every trip was submitted. No vehicle IDs. 3 of 7 months of 2026 published.
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11 FDE judgement call (demo): withhold Lyft's KPI instead of reporting either available number

A · Naive rate
92.7% → "compliant". Counts impossible negative times as on-time.

B · Drop bad rows
81.0% → "non-compliant", notice issued. Survivors are a biased slice if the request time is stamped late.

C · Impute
Fabricates the regulated quantity itself.

D · Withhold + bounds + data request ✓
NOT MEASURABLE (31–93%); names the field, base, record path and size.

Systematic, not noise: 58–61% in 3 months; one base; Lyft's WAV rows are its only rows with `originating_base_num` set (separate record path); no pre-booking signature; even "valid" rows are slower than every peer (P90 12.5 vs 8.5 min). **One rule for all:** bounds on one side of 90% → proven (Uber, Apr); straddling, with ≥ 90% clean → estimate (Uber, May/Jul; the line is a stated policy equal to the standard's tolerance); otherwise withheld (Lyft: bounds 31–93%, 40% clean).

Decisions: Lyft → data-correction request · Uber → compliant; confirming the pre-booked rides would make May/Jul proven · supply → Queens 20–24 (every month) + July's Queens/Bronx night windows · TLC → add a pre-booked flag and unserved-request records · June 2026 → hold.